Newman-Penrose Formalism Calculations for

metric:

Name: Stephani et al., Equation (21.30)

Reference(s): [1]

Line element:

$$ds^2 = -X\left(x_0, x_1, x_2, x_3\right) \mathrm{d}x_0 \otimes \mathrm{d}x_0 + \frac{R\left(x_0\right)^2}{X\left(x_0, x_1, x_2, x_3\right)} \mathrm{d}x_1 \otimes \mathrm{d}x_1 + \frac{R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2}{X\left(x_0, x_1, x_2, x_3\right)} \mathrm{d}x_2 \otimes \mathrm{d}x_2 + \frac{R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}{X\left(x_0, x_1, x_2, x_3\right)} \mathrm{d}x_3 \otimes \mathrm{d}x_3 \tag{1}$$

Coordinates: $\left(x_0, x_1, x_2, x_3\right)$

Null tetrad components:

Covariant components:

$$l = \left(\sqrt{\frac{1}{2}} \sqrt{X\left(x_0, x_1, x_2, x_3\right)}\right) dx_0 + \left(\frac{\sqrt{\frac{1}{2}} R\left(x_0\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_1 + (0)\, dx_2 + (0)\, dx_3 \tag{2}$$

$$n = \left(\sqrt{\frac{1}{2}} \sqrt{X\left(x_0, x_1, x_2, x_3\right)}\right) dx_0 + \left(-\frac{\sqrt{\frac{1}{2}} R\left(x_0\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_1 + (0)\, dx_2 + (0)\, dx_3 \tag{3}$$

$$m = (0)\, dx_0 + (0)\, dx_1 + \left(\frac{\sqrt{\frac{1}{2}} R\left(x_0\right) \Sigma\left(x_1, k\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_2 + \left(\frac{i \sqrt{\frac{1}{2}} R\left(x_0\right) \Sigma\left(x_1, k\right) \sin\left(x_2\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_3 \tag{4}$$

$$\overline{m} = (0)\, dx_0 + (0)\, dx_1 + \left(\frac{\sqrt{\frac{1}{2}} R\left(x_0\right) \Sigma\left(x_1, k\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_2 + \left(-\frac{i \sqrt{\frac{1}{2}} R\left(x_0\right) \Sigma\left(x_1, k\right) \sin\left(x_2\right)}{\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_3 \tag{5}$$

Contravariant components:

$$l = \left(-\frac{\sqrt{2}}{2\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_0 + \left(\frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)}\right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{6}$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}\right) dx_0 + \left(-\frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)}\right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{7}$$

$$m = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)}\right) dx_2 + \left(\frac{i\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)\sin\left(x_2\right)}\right) dx_3 \tag{8}$$

$$\overline{m} = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)}\right) dx_2 + \left(-\frac{i\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)\sin\left(x_2\right)}\right) dx_3 \tag{9}$$

Directional derivatives:

$$D = -\frac{\sqrt{2}\frac{\partial}{\partial x_0} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} + \frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_1} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)} \tag{10}$$

$$\Delta = -\frac{\sqrt{2}\frac{\partial}{\partial x_0} f\left(x_0, x_1, x_2, x_3\right)}{2\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} - \frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_1} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)} \tag{11}$$

$$\delta = \frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_2} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)} + \frac{i\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_3} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)\sin\left(x_2\right)} \tag{12}$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_2} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)} - \frac{i\sqrt{2}\sqrt{X\left(x_0, x_1, x_2, x_3\right)}\frac{\partial}{\partial x_3} f\left(x_0, x_1, x_2, x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1, k\right)\sin\left(x_2\right)} \tag{13}$$

Spin coefficients:

$$\rho = \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{14}$$

$$\sigma = 0 \tag{15}$$

$$\kappa = -\frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\,\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{16}$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{17}$$

$$\lambda = 0 \tag{18}$$

$$\nu = \frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\,\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{19}$$

$$\alpha = -\frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\,\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{20}$$

$$\beta = \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\,\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{21}$$

$$\tau = 0 \tag{22}$$

$$\pi = 0 \tag{23}$$

$$\epsilon = -\frac{\sqrt{2}\,\frac{\partial}{\partial x_0} R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} + \frac{\sqrt{2}\,\frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{8\,X\left(x_0, x_1, x_2, x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\,\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} \tag{24}$$

$$\gamma = \frac{\sqrt{2}\,\frac{\partial}{\partial x_0} R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} - \frac{\sqrt{2}\,\frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{8\,X\left(x_0, x_1, x_2, x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\,\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0, x_1, x_2, x_3\right)}} \tag{25}$$

Relations between spin coefficients:

Ricci tensor components:

$$\Phi_{00} = \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3}$$
$$+ \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{\frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)}$$
$$+ \frac{3 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3 \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} + \frac{\cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)}$$
$$- \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2} \tag{26}$$

$$\Phi_{01} = -\frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2}$$
$$- \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{i \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{i \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \tag{27}$$
$$+ \frac{3i \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} - \frac{i \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)}$$

$$\Phi_{02} = -\frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{i \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} \tag{28}$$

$$\Phi_{10} = -\frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2}$$
$$- \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{i\, \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{i\, \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \tag{29}$$
$$- \frac{3i\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} + \frac{i\, \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)}$$

$$\Phi_{11} = \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{4\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2}$$
$$- \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, X\left(x_0, x_1, x_2, x_3\right)^3} + \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{8\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{\frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} - \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \tag{30}$$
$$+ \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2} + \frac{X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} + \frac{\cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} - \frac{3\, \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)}$$
$$+ \frac{\frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{3\, \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}$$

$$\Phi_{12} = -\frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2}$$
$$+ \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{i\, \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{i\, \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \tag{31}$$
$$+ \frac{3i\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} + \frac{i\, \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)}$$

$$\Phi_{20} = -\frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{i\, \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} \tag{32}$$

$$
\begin{aligned}
\Phi_{21} = & -\frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& + \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{i\, \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{i\, \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{3i\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} - \frac{i\, \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)}
\end{aligned}
\tag{33}
$$

$$
\begin{aligned}
\Phi_{22} = & \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} \\
& + \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{\frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} \\
& - \frac{3\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} + \frac{\cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}
\end{aligned}
\tag{34}
$$

$$
\begin{aligned}
\Lambda = & \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{4\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right)^2}{12\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{6\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} \\
& - \frac{5\, \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{5\, \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, X\left(x_0, x_1, x_2, x_3\right)^3} - \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{8\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{\frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{12\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} \\
& - \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{24\, R\left(x_0\right)^2} + \frac{X\left(x_0, x_1, x_2, x_3\right)}{12\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} + \frac{\cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{24\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{24\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{16\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{24\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}
\end{aligned}
\tag{35}
$$

Weyl tensor components:

$$\Psi_0 = -\frac{\cos(x_2)\,\frac{\partial}{\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} + \frac{\frac{\partial^2}{(\partial x_2)^2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} + \frac{i\,\frac{\partial^2}{\partial x_2\partial x_3}X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} - \frac{i\,\cos(x_2)\,\frac{\partial}{\partial x_3}X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} \tag{36}$$

$$\Psi_1 = -\frac{\frac{\partial^2}{\partial x_1\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)} + \frac{\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{i\,\frac{\partial^2}{\partial x_1\partial x_3}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\sin(x_2)} + \frac{i\,\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_3}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} \tag{37}$$

$$\begin{aligned}\Psi_2 = {} & \frac{X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_1}\Sigma(x_1,k)^2}{6\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{X(x_0,x_1,x_2,x_3)\,\frac{\partial^2}{(\partial x_1)^2}\Sigma(x_1,k)}{6\,R(x_0)^2\,\Sigma(x_1,k)} - \frac{\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_1}X(x_0,x_1,x_2,x_3)}{6\,R(x_0)^2\,\Sigma(x_1,k)} + \frac{\frac{\partial^2}{(\partial x_1)^2}X(x_0,x_1,x_2,x_3)}{6\,R(x_0)^2} \\ & - \frac{X(x_0,x_1,x_2,x_3)}{6\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{\cos(x_2)\,\frac{\partial}{\partial x_2}X(x_0,x_1,x_2,x_3)}{12\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} - \frac{\frac{\partial^2}{(\partial x_2)^2}X(x_0,x_1,x_2,x_3)}{12\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X(x_0,x_1,x_2,x_3)}{12\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2}\end{aligned} \tag{38}$$

$$\Psi_3 = \frac{\frac{\partial^2}{\partial x_1\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)} - \frac{\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{i\,\frac{\partial^2}{\partial x_1\partial x_3}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\sin(x_2)} + \frac{i\,\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_3}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} \tag{39}$$

$$\Psi_4 = -\frac{\cos(x_2)\,\frac{\partial}{\partial x_2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} + \frac{\frac{\partial^2}{(\partial x_2)^2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{i\,\frac{\partial^2}{\partial x_2\partial x_3}X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} + \frac{i\,\cos(x_2)\,\frac{\partial}{\partial x_3}X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} \tag{40}$$

Newman-Penrose equations:

$$NP1 = 0 \tag{41}$$

$$NP2 = 0 \tag{42}$$

$$NP3 = 0 \tag{43}$$

$$NP4 = 0 \tag{44}$$

$$NP5 = 0 \tag{45}$$

$$NP6 = 0 \tag{46}$$

$$NP7 = 0 \tag{47}$$

$$NP8 = 0 \tag{48}$$

$$NP9 = 0 \tag{49}$$

$$NP10 = 0 \tag{50}$$

$$NP11 = 0 \tag{51}$$

$$NP12 = 0 \tag{52}$$

$$NP13 = 0 \tag{53}$$

$$NP14 = 0 \tag{54}$$

$$NP15 = 0 \tag{55}$$

$$NP16 = 0 \tag{56}$$

$$NP17 = 0 \tag{57}$$

$$NP18 = 0 \tag{58}$$

Bianchi identities:

$$BI1 = 0 \tag{59}$$

$$BI2 = 0 \tag{60}$$

$$BI3 = 0 \tag{61}$$

$$BI4 = 0 \tag{62}$$

$$BI5 = 0 \tag{63}$$

$$BI6 = 0 \tag{64}$$

$$BI7 = 0 \tag{65}$$

$$BI8 = 0 \tag{66}$$

$$BI9 = 0 \tag{67}$$

$$BI10 = 0 \tag{68}$$

$$BI11 = 0 \tag{69}$$

Petrov classification:

Petrov class:

A Type transformations:

Spin coefficients:

$$\rho' = \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{70}$$

$$\sigma' = -\frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{71}$$

$$\kappa' = \kappa \tag{72}$$

$$\mu' = -\frac{\sqrt{2}z^2\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z^2\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z^2\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{R\left(x_0\right)\Sigma\left(x_1,k\right)} - \frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{2\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{73}$$

$$\lambda' = -\frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{74}$$

$$\nu' = -\frac{\sqrt{2}z^2\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} + \frac{\sqrt{2}z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z^3\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^3\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{75}$$

$$\alpha' = \frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} - \frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}}$$
$$+\frac{3\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{76}$$
$$-\frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}$$

$$\beta' = -\frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}$$
$$-\frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} + \frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} \tag{77}$$
$$-\frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}$$

$$\tau' = -\frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{78}$$
$$-\frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}$$

$$\pi' = -\frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{79}$$

$$\epsilon' = -\frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{80}$$
$$-\frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}$$

$$
\begin{aligned}
\gamma' = & -\frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} R(x_0)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} + \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{8 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} + \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} z^2 \overline{z} \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} \\
& + \frac{\sqrt{2} \overline{z} \frac{\partial}{\partial x_0} R(x_0)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} \sqrt{X(x_0, x_1, x_2, x_3)} \overline{z} \frac{\partial}{\partial x_1} \Sigma(x_1, k)}{2 R(x_0) \Sigma(x_1, k)} - \frac{\sqrt{2} \overline{z} \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{8 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} + \frac{3 \sqrt{2} \overline{z} \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} \\
& - \frac{i \sqrt{2} z^2 \overline{z} \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} + \frac{\sqrt{2} z \sqrt{X(x_0, x_1, x_2, x_3)} \cos(x_2)}{4 R(x_0) \Sigma(x_1, k) \sin(x_2)} + \frac{\sqrt{2} \frac{\partial}{\partial x_0} R(x_0)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{8 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} \\
& + \frac{\sqrt{2} \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} z \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} z \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)}
\end{aligned}
\tag{81}
$$

Weyl components:

$$
\Psi_0' = \Psi_0 \tag{82}
$$

$$
\begin{aligned}
\Psi_1' = & -\frac{\frac{\partial^2}{\partial x_1 \partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)} - \frac{z \cos(x_2) \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} + \frac{\frac{\partial}{\partial x_1} \Sigma(x_1, k) \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2} + \frac{z \frac{\partial^2}{(\partial x_2)^2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2} - \frac{i \frac{\partial^2}{\partial x_1 \partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k) \sin(x_2)} \\
& + \frac{i z \frac{\partial^2}{\partial x_2 \partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} - \frac{i z \cos(x_2) \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)^2} + \frac{i \frac{\partial}{\partial x_1} \Sigma(x_1, k) \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} - \frac{z \frac{\partial^2}{(\partial x_3)^2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)^2}
\end{aligned}
\tag{83}
$$

$$
\begin{aligned}
\Psi_2' = & \frac{X(x_0, x_1, x_2, x_3) \frac{\partial}{\partial x_1} \Sigma(x_1, k)^2}{6 R(x_0)^2 \Sigma(x_1, k)^2} - \frac{X(x_0, x_1, x_2, x_3) \frac{\partial^2}{(\partial x_1)^2} \Sigma(x_1, k)}{6 R(x_0)^2 \Sigma(x_1, k)} - \frac{\frac{\partial}{\partial x_1} \Sigma(x_1, k) \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{6 R(x_0)^2 \Sigma(x_1, k)} + \frac{\frac{\partial^2}{(\partial x_1)^2} X(x_0, x_1, x_2, x_3)}{6 R(x_0)^2} \\
& - \frac{z \frac{\partial^2}{\partial x_1 \partial x_2} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)} - \frac{z^2 \cos(x_2) \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} + \frac{z \frac{\partial}{\partial x_1} \Sigma(x_1, k) \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2} + \frac{z^2 \frac{\partial^2}{(\partial x_2)^2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2} \\
& - \frac{i z \frac{\partial^2}{\partial x_1 \partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k) \sin(x_2)} + \frac{i z^2 \frac{\partial^2}{\partial x_2 \partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} - \frac{i z^2 \cos(x_2) \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)^2} + \frac{i z \frac{\partial}{\partial x_1} \Sigma(x_1, k) \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{2 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} \\
& - \frac{X(x_0, x_1, x_2, x_3)}{6 R(x_0)^2 \Sigma(x_1, k)^2} - \frac{\cos(x_2) \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{12 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)} - \frac{\frac{\partial^2}{(\partial x_2)^2} X(x_0, x_1, x_2, x_3)}{12 R(x_0)^2 \Sigma(x_1, k)^2} - \frac{z^2 \frac{\partial^2}{(\partial x_3)^2} X(x_0, x_1, x_2, x_3)}{4 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2} X(x_0, x_1, x_2, x_3)}{12 R(x_0)^2 \Sigma(x_1, k)^2 \sin(x_2)^2}
\end{aligned}
\tag{84}
$$

$$
\begin{aligned}
\Psi_3' = {} & \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2} \\
& - \frac{3\,z^2\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^3\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{3\,z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z^3\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{3i\,z^2\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z^3\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{i\,z^3\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{3i\,z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& - \frac{zX\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{z^3\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{85}
$$

$$
\begin{aligned}
\Psi_4' = {} & \frac{z^2X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2} \\
& - \frac{z^3\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^4\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z^4\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{i\,z^3\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z^4\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{i\,z^4\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{i\,z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& - \frac{z^2X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{z^4\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,z\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& + \frac{\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{i\,\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{86}
$$

Ricci components:

$$
\Phi_{00}' = \Phi_{00} \tag{87}
$$

$$
\begin{aligned}
\Phi'_{01} = & \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} \\
& + \frac{z \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{z \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} \\
& + \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& - \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} \\
& - \frac{i \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{i \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{3i \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} \\
& - \frac{i \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{z \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{z \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}
\end{aligned}
\tag{88}
$$

$$
\begin{aligned}
\Phi'_{02} = {} & \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& - \frac{3\, z^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} + \frac{z^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{z^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z^2 \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{3\, z^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z^2 \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{z \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{z^2 \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& - \frac{z^2 \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{i\, z \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{i\, z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{3 i\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} - \frac{i\, z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{z^2 \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2} \\
& - \frac{i \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2}
\end{aligned}
\tag{89}
$$

$$
\begin{aligned}
\Phi'_{10} = & \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} \\
& + \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{\overline{z}\frac{\partial^2}{\partial x_0\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} \\
& + \frac{3\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} - \frac{\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& + \frac{i\,\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{90}
$$

$$
\begin{aligned}
\Phi'_{11} = {} & \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z\overline{z}\frac{\partial^2}{\partial x_0\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{4\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\overline{z}\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2} \\
& + \frac{z\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,z\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,\overline{z}\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{i\,\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{3i\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& + \frac{i\,z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{3\,\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{3\,\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{91}
$$

$$
\begin{aligned}
\Phi'_{12} = {} & \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}\frac{\partial^2}{\partial x_0\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} \\
& + \frac{z^2\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^2\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z^2\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,z^2\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z^2\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{i\,z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,z^2\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{3i\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& + \frac{i\,z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{zX\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{3\,z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} \\
& + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)}
\end{aligned}
\tag{92}
$$

$$
\begin{aligned}
\Phi'_{20} = {} & \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& - \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{\overline{z}^2 \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{\overline{z} \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{\overline{z}^2 \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{\overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, \overline{z} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{\overline{z} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& - \frac{\overline{z}^2 \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} + \frac{i\, \overline{z} \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& + \frac{i\, \overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{3 i\, \overline{z} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} + \frac{i\, \overline{z} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\overline{z}^2 \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2} \\
& + \frac{i\, \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2}
\end{aligned}
\tag{93}
$$

$$
\begin{aligned}
\Phi'_{21} = {} & \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)^2}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} - \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R(x_0)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} - \frac{zX(x_0,x_1,x_2,x_3)\,\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2}\Sigma(x_1,k)}{2\,R(x_0)^2\,\Sigma(x_1,k)} + \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} \\
& - \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)^2}{8\,X(x_0,x_1,x_2,x_3)^3} + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X(x_0,x_1,x_2,x_3)}{4\,X(x_0,x_1,x_2,x_3)^2} - \frac{z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{z\overline{z}^2 \frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)} + \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} - \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2} + \frac{\overline{z}\frac{\partial}{\partial x_0} R(x_0)^2}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} - \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2} R(x_0)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} - \frac{X(x_0,x_1,x_2,x_3)\,\overline{z}\frac{\partial}{\partial x_1}\Sigma(x_1,k)^2}{2\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& + \frac{\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} - \frac{3\,\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)^2}{8\,X(x_0,x_1,x_2,x_3)^3} + \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2} X(x_0,x_1,x_2,x_3)}{4\,X(x_0,x_1,x_2,x_3)^2} - \frac{z\overline{z}\frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& - \frac{\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)} - \frac{\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_1)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2} \\
& + \frac{z\overline{z}^2 \cos(x_2)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2 \sin(x_2)} - \frac{z\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{3\,z\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} + \frac{3\,\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} - \frac{z\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& - \frac{\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& + \frac{i\,z\overline{z}\frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} - \frac{i\,\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{i\,z\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} \\
& - \frac{i\,\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} - \frac{3i\,z\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2\sin(x_2)} + \frac{3i\,\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2\sin(x_2)} \\
& + \frac{i\,z\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} - \frac{i\,\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{X(x_0,x_1,x_2,x_3)\,\overline{z}}{2\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& - \frac{\frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\cos(x_2)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} - \frac{\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{3\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} + \frac{\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{z\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} \\
& - \frac{3\,\overline{z}\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_2)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)^2}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)\sin(x_2)^2} \\
& + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_3)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} + \frac{i\,\frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{i\,\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)}
\end{aligned}
\tag{94}
$$

$$
\begin{aligned}
\Phi'_{22} = {} & \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z X\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{z\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{2\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2} \\
& + \frac{z^2\overline{z}^2\cos\left(x_2\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^2\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,z^2\overline{z}\frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z^2\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,z^2\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{3i\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& + \frac{i\,z^2\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{z X\left(x_0,x_1,x_2,x_3\right)\overline{z}}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} \\
& + \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{\frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{3\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& \frac{3\,z\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \quad \frac{3\,\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \quad \frac{z\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{}
\end{aligned}
$$

$$\Lambda' = \Lambda \tag{96}$$

B Type transformations:

Spin coefficients:

$$\rho' = \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right)}{2\, R\left(x_0\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)}} - \frac{\sqrt{2} z \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \overline{z} \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right)} - \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)}} \tag{97}$$

$$\sigma' = 0 \tag{98}$$

$$\kappa' = -\frac{\sqrt{2} z^3 \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)}} - \frac{i \sqrt{2} z^3 \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \sin\left(x_2\right)} \tag{99}$$

$$\mu' = -\frac{\sqrt{2} \frac{\partial}{\partial x_0} R\left(x_0\right)}{2\, z R\left(x_0\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \overline{z}} - \frac{\sqrt{2} \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right)}{2\, z R\left(x_0\right) \Sigma\left(x_1, k\right) \overline{z}} + \frac{\sqrt{2} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, z X\left(x_0, x_1, x_2, x_3\right)^{\frac{3}{2}} \overline{z}} + \frac{\sqrt{2} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, z R\left(x_0\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \overline{z}} \tag{100}$$

$$\lambda' = 0 \tag{101}$$

$$\nu' = \frac{\sqrt{2} \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, z^3 R\left(x_0\right) \Sigma\left(x_1, k\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)}} - \frac{i \sqrt{2} \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, z^3 R\left(x_0\right) \Sigma\left(x_1, k\right) \sqrt{X\left(x_0, x_1, x_2, x_3\right)} \sin\left(x_2\right)} \tag{102}$$

$$\alpha' = \alpha \tag{103}$$

$$\beta' = \beta \tag{104}$$

$$\tau' = 0 \tag{105}$$

$$\pi' = 0 \tag{106}$$

$$\epsilon' = -\frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} \tag{107}$$

$$\gamma' = \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,zR\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,zX\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}\,\overline{z}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,zR\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}} \tag{108}$$

Weyl components:

$$\begin{aligned}\Psi_0' = &-\frac{z^4\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{z^4\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{i\,z^4\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\ &- \frac{i\,z^4\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{z^4\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}\end{aligned} \tag{109}$$

$$\Psi_1' = -\frac{z^3\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{i\,z^3\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \tag{110}$$

$$\begin{aligned}\Psi_2' = &\frac{z^2X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2} \\ &- \frac{z^2X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^2\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}\end{aligned} \tag{111}$$

$$\Psi_3' = \frac{z\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{i\,z\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \tag{112}$$

$$\Psi_4' = \Psi_4 \tag{113}$$

Ricci components:

$$
\begin{aligned}
\Phi'_{00} = {} & \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right) \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z^2\overline{z}^2 \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 \sin\left(x_2\right)} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_3)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{114}
$$

$$
\begin{aligned}
\Phi'_{01} = {} & -\frac{z^2\overline{z} \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,z^2\overline{z} \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{z^2\overline{z} \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{i\,z^2\overline{z} \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z^2\overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& + \frac{3i\,z^2\overline{z} \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} - \frac{i\,z^2\overline{z} \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)}
\end{aligned}
\tag{115}
$$

$$
\Phi'_{02} = -\frac{z^2 \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{i\,z^2 \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{z^2 \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2}
\tag{116}
$$

$$
\begin{aligned}
\Phi'_{10} = {} & -\frac{z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{i\,z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{3i\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right) \frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)}
\end{aligned}
\tag{117}
$$

$$
\begin{aligned}
\Phi'_{11} = {} & \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{4\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2} + \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{118}
$$

$$
\begin{aligned}
\Phi'_{12} = {} & -\frac{z\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)^2} \\
& + \frac{z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} - \frac{i\,z\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& + \frac{3i\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)}
\end{aligned}
\tag{119}
$$

$$
\Phi'_{20} = -\frac{\overline{z}^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{i\,\overline{z}^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{\overline{z}^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2}
\tag{120}
$$

$$
\begin{aligned}
\Phi'_{21} = {} & -\frac{\overline{z}\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)^2} \\
& + \frac{\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)} + \frac{i\,\overline{z}\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{3i\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} - \frac{i\,\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)}
\end{aligned}
\tag{121}
$$

$$\Phi'_{22} = \Phi_{22} \tag{122}$$

$$\Lambda' = \Lambda \tag{123}$$

C Type transformations:

Spin coefficients:

$$\begin{aligned}\rho' = &-\frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)}-\frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)}\\
&+\frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}}+\frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}}+\frac{\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&+\frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{\sqrt{2}\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)}\\
&-\frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}}+\frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{i\sqrt{2}z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}\end{aligned}\tag{124}$$

$$\begin{aligned}\sigma' = &-\frac{\sqrt{2}z^2\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)}+\frac{\sqrt{2}z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}z^3\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&-\frac{i\sqrt{2}z^3\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}+\frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)}\\
&-\frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}\end{aligned}\tag{125}$$

$$\begin{aligned}\kappa' = &-\frac{\sqrt{2}z^2\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)}+\frac{\sqrt{2}z^2\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}z^3\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&-\frac{i\sqrt{2}z^3\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}-\frac{\sqrt{2}z^2\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)}+\frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)}\\
&-\frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}}+\frac{\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}+\frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&-\frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}-\frac{i\sqrt{2}z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}\\
&-\frac{\sqrt{2}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}-\frac{i\sqrt{2}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}\end{aligned}\tag{126}$$

$$\mu' = \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} R(x_0)}{2 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{4 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} + \frac{\sqrt{2} z \overline{z} \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} + \frac{\sqrt{2} z^2 \overline{z} \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}}$$
$$- \frac{i \sqrt{2} z^2 \overline{z} \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} - \frac{\sqrt{2} z \sqrt{X(x_0, x_1, x_2, x_3)} \cos(x_2)}{2 R(x_0) \Sigma(x_1, k) \sin(x_2)} - \frac{\sqrt{2} \frac{\partial}{\partial x_0} R(x_0)}{2 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} \sqrt{X(x_0, x_1, x_2, x_3)} \frac{\partial}{\partial x_1} \Sigma(x_1, k)}{2 R(x_0) \Sigma(x_1, k)}$$
$$+ \frac{\sqrt{2} \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{4 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} + \frac{\sqrt{2} \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} + \frac{\sqrt{2} z \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} z \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} \tag{127}$$

$$\lambda' = \frac{\sqrt{2} z \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} z \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} \tag{128}$$

$$\nu' = \nu \tag{129}$$

$$\alpha' = \frac{\sqrt{2} z \frac{\partial}{\partial x_0} R(x_0)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} z \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{8 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}} + \frac{\sqrt{2} z \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}}$$
$$+ \frac{\sqrt{2} z^2 \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} z^2 \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} - \frac{\sqrt{2} \sqrt{X(x_0, x_1, x_2, x_3)} \cos(x_2)}{4 R(x_0) \Sigma(x_1, k) \sin(x_2)} \tag{130}$$
$$+ \frac{\sqrt{2} \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)}$$

$$\beta' = - \frac{\sqrt{2} z \frac{\partial}{\partial x_0} R(x_0)}{4 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{\sqrt{2} z \sqrt{X(x_0, x_1, x_2, x_3)} \frac{\partial}{\partial x_1} \Sigma(x_1, k)}{2 R(x_0) \Sigma(x_1, k)} + \frac{\sqrt{2} z \frac{\partial}{\partial x_0} X(x_0, x_1, x_2, x_3)}{8 X(x_0, x_1, x_2, x_3)^{\frac{3}{2}}}$$
$$+ \frac{3 \sqrt{2} z \frac{\partial}{\partial x_1} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \sqrt{X(x_0, x_1, x_2, x_3)}} + \frac{\sqrt{2} z^2 \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} z^2 \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{4 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)} \tag{131}$$
$$+ \frac{\sqrt{2} \sqrt{X(x_0, x_1, x_2, x_3)} \cos(x_2)}{4 R(x_0) \Sigma(x_1, k) \sin(x_2)} - \frac{\sqrt{2} \frac{\partial}{\partial x_2} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)}} - \frac{i \sqrt{2} \frac{\partial}{\partial x_3} X(x_0, x_1, x_2, x_3)}{8 R(x_0) \Sigma(x_1, k) \sqrt{X(x_0, x_1, x_2, x_3)} \sin(x_2)}$$

$$\tau' = \frac{\sqrt{2}z\frac{\partial}{\partial x_0}R\left(x_0\right)}{2\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{132}$$

$$\pi' = \frac{\sqrt{2}z^2\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z^2\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{133}$$

$$\begin{aligned}\epsilon' = &-\frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right)} + \frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{3\sqrt{2}z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&+\frac{\sqrt{2}z^2\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\overline{z}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{\sqrt{2}z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&-\frac{i\sqrt{2}z^2\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} - \frac{\sqrt{2}z\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\cos\left(x_2\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&+\frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}}\\
&-\frac{i\sqrt{2}z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)}\end{aligned} \tag{134}$$

$$\gamma' = \frac{\sqrt{2}\frac{\partial}{\partial x_0}R\left(x_0\right)}{4\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{\sqrt{2}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^{\frac{3}{2}}} + \frac{\sqrt{2}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} + \frac{\sqrt{2}z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}} - \frac{i\sqrt{2}z\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right)\sqrt{X\left(x_0,x_1,x_2,x_3\right)}\sin\left(x_2\right)} \tag{135}$$

Weyl components:

$$
\begin{aligned}
\Psi_0' = {} & \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2} \\
& + \frac{z^3\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^4\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z^4\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{i\,z^3\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{i\,z^4\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{i\,z^4\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{i\,z^3\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{z^4\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,z\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& + \frac{\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{i\,\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{i\,\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{136}
$$

$$
\begin{aligned}
\Psi_1' = {} & \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2} \\
& + \frac{3\,z^2\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^3\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{3\,z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z^3\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{3i\,z^2\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{i\,z^3\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{i\,z^3\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{3i\,z^2\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} \\
& - \frac{zX\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{z^3\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\,\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} + \frac{i\,\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{137}
$$

$$\Psi_2' = \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2}$$
$$+ \frac{z\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z^2\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z^2\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \tag{138}$$
$$- \frac{i\,z\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)} - \frac{i\,z^2\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{i\,z^2\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{i\,z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)}$$
$$- \frac{X\left(x_0,x_1,x_2,x_3\right)}{6\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{12\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}$$

$$\Psi_3' = \frac{\frac{\partial^2}{\partial x_1\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{z\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{i\,\frac{\partial^2}{\partial x_1\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)\sin\left(x_2\right)}$$
$$- \frac{i\,z\frac{\partial^2}{\partial x_2\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} + \frac{i\,z\cos\left(x_2\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} + \frac{i\,\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} \tag{139}$$

$$\Psi_4' = \Psi_4 \tag{140}$$

Ricci components:

$$
\begin{aligned}
\Phi'_{00} = {} & \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z^2\overline{z}^2 \frac{\partial^2}{\partial x_0\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z^2\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{z\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{2\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\overline{z}\frac{\partial^2}{\partial x_0\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}^2 \frac{\partial^2}{\partial x_0\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2} \\
& + \frac{z^2\overline{z}^2\cos\left(x_2\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^2\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z^2\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}^2 \frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{i\,z^2\overline{z}\frac{\partial^2}{\partial x_0\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z\overline{z}^2 \frac{\partial^2}{\partial x_0\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z^2\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& + \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{3i\,z^2\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} - \frac{3i\,z\overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& + \frac{i\,z^2\overline{z}\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}}{R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} \\
& + \frac{\frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{\frac{\partial^2}{\partial x_0\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\frac{\partial^2}{\partial x_0\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial^2}{\partial x_0\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{3\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& \frac{3\,z\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{\ldots} \quad \frac{3\,\overline{z}\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{\ldots} \quad \frac{z\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{\ldots}
\end{aligned}
$$

$$
\begin{aligned}
\Phi'_{01} = {} & \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2 X\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z^2\overline{z}\frac{\partial^2}{\partial x_0\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z^2\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z^2\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{z\overline{z}\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} \\
& + \frac{z^2\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z^2\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z^2\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z^2\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,z^2\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,z\overline{z}\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z^2\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{i\,z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,z^2\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{3i\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& - \frac{i\,z^2\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{zX\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& - \frac{\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{3\,z\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z^2\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} \\
& + \frac{z^2\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{i\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& \frac{3i\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{} \quad \frac{i\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{} \quad \frac{i\,\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{}
\end{aligned}
$$

$$
\begin{aligned}
\Phi'_{02} = {} & \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& - \frac{3\, z^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} + \frac{z^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{z^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{z^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z^2 \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} - \frac{3\, z^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z^2 \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{z \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{z^2 \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& - \frac{z^2 \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} - \frac{i\, z \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{i\, z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{3 i\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} + \frac{i\, z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z^2 \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{z^2 \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2} \\
& - \frac{i \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2}
\end{aligned}
\tag{143}
$$

$$
\begin{aligned}
\Phi'_{10} = {} & \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)^2}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} - \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R(x_0)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} - \frac{zX(x_0,x_1,x_2,x_3)\,\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2}\Sigma(x_1,k)}{2\,R(x_0)^2\,\Sigma(x_1,k)} + \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} \\
& - \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)^2}{8\,X(x_0,x_1,x_2,x_3)^3} + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X(x_0,x_1,x_2,x_3)}{4\,X(x_0,x_1,x_2,x_3)^2} + \frac{z\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} + \frac{z\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{z\overline{z}^2 \frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)} - \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} - \frac{3\,z\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2} + \frac{\overline{z}\frac{\partial}{\partial x_0} R(x_0)^2}{2\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} - \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2} R(x_0)}{2\,R(x_0)\,X(x_0,x_1,x_2,x_3)} - \frac{X(x_0,x_1,x_2,x_3)\,\overline{z}\frac{\partial}{\partial x_1}\Sigma(x_1,k)^2}{2\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& + \frac{\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,X(x_0,x_1,x_2,x_3)^2} - \frac{3\,\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)^2}{8\,X(x_0,x_1,x_2,x_3)^3} + \frac{\overline{z}\frac{\partial^2}{(\partial x_0)^2} X(x_0,x_1,x_2,x_3)}{4\,X(x_0,x_1,x_2,x_3)^2} - \frac{z\overline{z}\frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& - \frac{\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial}{\partial x_1}\Sigma(x_1,k)\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)} - \frac{\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_1)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2} \\
& + \frac{z\overline{z}^2 \cos(x_2)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2 \sin(x_2)} - \frac{z\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{3\,z\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} + \frac{3\,\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} + \frac{z\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& + \frac{i\,z\overline{z}\frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} - \frac{i\,\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{i\,z\overline{z}\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{2\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} \\
& - \frac{i\,\overline{z}^2 \frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} - \frac{3i\,z\overline{z}\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2\sin(x_2)} + \frac{3i\,\overline{z}^2 \frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2\sin(x_2)} \\
& - \frac{i\,z\overline{z}\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{i\,\overline{z}^2 \frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{X(x_0,x_1,x_2,x_3)\,\overline{z}}{2\,R(x_0)^2\,\Sigma(x_1,k)^2} \\
& - \frac{\frac{\partial^2}{\partial x_0 \partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\cos(x_2)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)} - \frac{\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} \\
& + \frac{3\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)^2} - \frac{\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)}{8\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)} - \frac{z\,\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} \\
& - \frac{3\,\overline{z}\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)^2}{8\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)} + \frac{\overline{z}\frac{\partial^2}{(\partial x_2)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2} - \frac{z\overline{z}^2 \frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)^2}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\,X(x_0,x_1,x_2,x_3)\sin(x_2)^2} \\
& + \frac{z\overline{z}^2 \frac{\partial^2}{(\partial x_3)^2} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)^2\sin(x_2)^2} + \frac{i\,\frac{\partial^2}{\partial x_0 \partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} + \frac{i\,\frac{\partial}{\partial x_0} R(x_0)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)}{4\,R(x_0)^2\,\Sigma(x_1,k)\,X(x_0,x_1,x_2,x_3)\sin(x_2)} \\
& 3i\,\frac{\partial}{\partial x_0} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3) \qquad i\,\frac{\partial}{\partial x_1} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3) \qquad i\,z\frac{\partial}{\partial x_2} X(x_0,x_1,x_2,x_3)\,\frac{\partial}{\partial x_3} X(x_0,x_1,x_2,x_3)
\end{aligned}
$$

$$
\begin{aligned}
\Phi'_{11} = {} & \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{zX\left(x_0,x_1,x_2,x_3\right)\overline{z}\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z\overline{z}\frac{\partial^2}{\partial x_0\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,z\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} + \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)^2}{4\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{\frac{\partial^2}{(\partial x_0)^2}R\left(x_0\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,X\left(x_0,x_1,x_2,x_3\right)^3} + \frac{\frac{\partial^2}{(\partial x_0)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{z\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& - \frac{\overline{z}\frac{\partial^2}{\partial x_0\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} - \frac{\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial^2}{(\partial x_1)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2} \\
& + \frac{z\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{3\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{3\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{z\overline{z}\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{z\overline{z}\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,z\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{i\,\overline{z}\frac{\partial^2}{\partial x_0\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,z\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} \\
& - \frac{i\,\overline{z}\frac{\partial}{\partial x_0}R\left(x_0\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,z\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} + \frac{3i\,\overline{z}\frac{\partial}{\partial x_0}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& - \frac{i\,z\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,\overline{z}\frac{\partial}{\partial x_1}X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{\cos\left(x_2\right)\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{3\,\frac{\partial}{\partial x_2}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\frac{\partial^2}{(\partial x_2)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} - \frac{z\overline{z}\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} \\
& + \frac{z\overline{z}\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2} - \frac{3\,\frac{\partial}{\partial x_3}X\left(x_0,x_1,x_2,x_3\right)^2}{16\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{\frac{\partial^2}{(\partial x_3)^2}X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{145}
$$

$$
\begin{aligned}
\Phi'_{12} = & \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{z \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z X\left(x_0, x_1, x_2, x_3\right) \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} \\
& + \frac{z \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{z \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} \\
& - \frac{3\, z \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, z \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{z \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& + \frac{\frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{z \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{z \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} \\
& - \frac{i \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{i \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{3i \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} \\
& + \frac{i \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{z \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{z \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2}
\end{aligned}
\tag{146}
$$

$$
\begin{aligned}
\Phi'_{20} = {} & \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} - \frac{X\left(x_0, x_1, x_2, x_3\right) \overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} \Sigma\left(x_1, k\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} + \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} \\
& - \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, X\left(x_0, x_1, x_2, x_3\right)^3} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{\overline{z}^2 \frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{\overline{z}^2 \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{\overline{z}^2 \frac{\partial}{\partial x_1} \Sigma\left(x_1, k\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)} - \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) X\left(x_0, x_1, x_2, x_3\right)^2} - \frac{3\, \overline{z}^2 \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 X\left(x_0, x_1, x_2, x_3\right)} \\
& + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_1)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2} - \frac{\overline{z} \frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{\overline{z}^2 \cos\left(x_2\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)} \\
& - \frac{\overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} + \frac{3\, \overline{z} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2} + \frac{\overline{z} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)} \\
& - \frac{\overline{z}^2 \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_2)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2} + \frac{i\, \overline{z} \frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& + \frac{i\, \overline{z} \frac{\partial}{\partial x_0} R\left(x_0\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{2\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} - \frac{3i\, \overline{z} \frac{\partial}{\partial x_0} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right) \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right)^2 \sin\left(x_2\right)} - \frac{i\, \overline{z} \frac{\partial}{\partial x_1} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right) X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} \\
& - \frac{\frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right)} - \frac{\overline{z}^2 \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2} + \frac{\overline{z}^2 \frac{\partial^2}{(\partial x_3)^2} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 \sin\left(x_2\right)^2} \\
& + \frac{i\, \frac{\partial}{\partial x_2} X\left(x_0, x_1, x_2, x_3\right) \frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)}{4\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)} + \frac{\frac{\partial}{\partial x_3} X\left(x_0, x_1, x_2, x_3\right)^2}{8\, R\left(x_0\right)^2 \Sigma\left(x_1, k\right)^2 X\left(x_0, x_1, x_2, x_3\right) \sin\left(x_2\right)^2}
\end{aligned}
\tag{147}
$$

$$
\begin{aligned}
\Phi'_{21} = & \frac{\overline{z}\,\frac{\partial}{\partial x_0} R\left(x_0\right)^2}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\,\frac{\partial^2}{(\partial x_0)^2} R\left(x_0\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{X\left(x_0,x_1,x_2,x_3\right)\overline{z}\,\frac{\partial^2}{(\partial x_1)^2}\Sigma\left(x_1,k\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} + \frac{\overline{z}\,\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\overline{z}\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,X\left(x_0,x_1,x_2,x_3\right)^3} \\
& + \frac{\overline{z}\,\frac{\partial^2}{(\partial x_0)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,X\left(x_0,x_1,x_2,x_3\right)^2} + \frac{\overline{z}\,\frac{\partial^2}{\partial x_0 \partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\,\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\,\frac{\partial}{\partial x_1}\Sigma\left(x_1,k\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{2\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)} \\
& - \frac{3\,\overline{z}\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right) X\left(x_0,x_1,x_2,x_3\right)^2} - \frac{3\,\overline{z}\,\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)^2}{8\,R\left(x_0\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\,\frac{\partial^2}{(\partial x_1)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2} - \frac{\frac{\partial^2}{\partial x_0 \partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} \\
& + \frac{\overline{z}\cos\left(x_2\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)} - \frac{\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} + \frac{3\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2} \\
& + \frac{\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)} - \frac{\overline{z}\,\frac{\partial}{\partial x_2} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)} + \frac{\overline{z}\,\frac{\partial^2}{(\partial x_2)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2} \\
& + \frac{i\,\frac{\partial^2}{\partial x_0 \partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} + \frac{i\,\frac{\partial}{\partial x_0} R\left(x_0\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{3i\,\frac{\partial}{\partial x_0} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)^2\sin\left(x_2\right)} \\
& - \frac{i\,\frac{\partial}{\partial x_1} X\left(x_0,x_1,x_2,x_3\right)\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)}{8\,R\left(x_0\right)^2\Sigma\left(x_1,k\right) X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)} - \frac{\overline{z}\,\frac{\partial}{\partial x_3} X\left(x_0,x_1,x_2,x_3\right)^2}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2 X\left(x_0,x_1,x_2,x_3\right)\sin\left(x_2\right)^2} + \frac{\overline{z}\,\frac{\partial^2}{(\partial x_3)^2} X\left(x_0,x_1,x_2,x_3\right)}{4\,R\left(x_0\right)^2\Sigma\left(x_1,k\right)^2\sin\left(x_2\right)^2}
\end{aligned}
\tag{148}
$$

$$
\Phi'_{22} = \Phi_{22} \tag{149}
$$

$$
\Lambda' = \Lambda \tag{150}
$$

References

[1] H. Stephani, D. Kramer, M. MacCallum, C. Hoenselaers, and E. Herlt, Exact Solutions of Einstein's Field Equations. Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2009.

Calculated by:

Name:

E-mail:

Date: